

## Fill Loops — English Worksheet

**Topic:** One corner, six rectangles · **Course:** 3D Coordinates · **Level:** Extension · **Time:** about 40 minutes

TYPE **DEL** BEFORE EVERY **RUN**. EVERY TIME.

Every **???** is one short thing for you to work out. There are no answers on this sheet.

### 1 · Read the loop 🔍

```
x_start = 1
z_start = 1
x_end = 1
z_end = 1
for i in range(1, 7):
    blocks.fill(EMERALD_BLOCK,
                pos(x_start, 0, z_start),
                pos(x_end, 0, z_end), REPLACE)
    x_end = x_end + 2
    z_end = z_end + 3
```

**range(1, 7)** gives you which numbers? Write them all.

|  |
|--|
|  |
|  |
|  |

Two lines inside the loop change a variable. Write them.

|  |
|--|
|  |
|  |
|  |

Four variables never change. Which?

|  |
|--|
|  |
|  |
|  |

**GOT IT RIGHT IF**

your list of numbers stops at 6, not 7.

2 · Fill the table 

Do it in your head first. Run the code after.

| I | X_END | Z_END | SIZE  | BLOCKS |
|---|-------|-------|-------|--------|
| 1 | 1     | 1     | 1 x 1 | 1      |
| 2 | 3     | 4     | 3 x 4 | 12     |
| 3 |       |       |       |        |
| 4 |       |       |       |        |
| 5 |       |       |       |        |
| 6 |       |       |       |        |

Add up the last column. How many blocks does the code write?

Now walk on your build and count it. How many blocks are really there?

The two numbers are different.

Why?

### 3 · One fill instead of six

Only the last rectangle is still visible.

#### TEMPLATE

```
bblocks.fill(EMERALD_BLOCK,  
    pos(1, 0, 1),  
    pos(???, 0, ???), REPLACE)
```

#### STEPS

1. Type `del` .
2. Run your one-line version.
3. Type `del` . Run the loop version.
4. Look at both. Same shape?

#### WORKS WHEN

both builds look exactly the same.

## 4 · Make it stand up

Right now `y_start` and `y_end` are both `0`, so your build is one block tall.

### TEMPLATE · INSIDE THE LOOP

```
x_end = x_end + 2  
z_end = z_end + 3  
??? = ??? + ???
```

#### START SMALL

Use **1** first. Adding 3 six times makes it 15 tall and your `del` cannot reach the top.

**Before you run it: how tall will it be after 6 turns if you add 1 each time?**

|  |
|--|
|  |
|  |
|  |

#### WORKS WHEN

it is a solid block you can stand next to, not a floor.

## 5 · Pull them apart ✂

The near corner never moves, so the shapes sit on top of each other. Move it and they separate.

TEMPLATE · INSIDE THE LOOP

```
??? = ??? + ???
```

**Tree of thought: tree 3 is 5 wide. How far must the corner move so shapes do not touch?**

|  |
|--|
|  |
|  |
|  |

### WORKS WHEN

you can walk between the shapes.

### STILL ONE LUMP?

You changed `x_end` again instead of `x_start`. Read the variable name carefully.

## 6 · Show Your Work 📷 🎥

Record **one video**, one take, no stopping. A phone is fine. Show these in order:

1. Type `del` , then `run` . Walk around your build.
2. Show the two lines inside the loop and say what each one does.
3. Show your filled table and say why the two totals are different.
4. Show your one-line version next to the loop version.
5. Show your standing-up build and say which variable you changed.

Try to use these words: **loop**, **variable**, **corner**, **fill**, **cover**.

**Plan what you will say before you record.**

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Submit** ✓

Send this worksheet and your video to teacher on KakaoTalk.